

Enhanced Crop Recommendation Using Machine Learning: A Comparative Study of Multiple Classification Algorithms

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Abstract—Accurate crop recommendation is essential for improving agricultural productivity and supporting informed decisionmaking by farmers. This paper presents a comprehensive evaluation of seven machine learning classification algorithms applied to the Crop Recommendation Dataset, which encompasses 2,200 instances described by seven soil and climatic parameters: nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall. The algorithms evaluated include Logistic Regression (LR), Gaussian Naïve Bayes (NB), K-Nearest Neighbors (KNN), Decision Tree (DT), Random Forest (RF), Extreme Gradient Boosting (XGBoost), and a multi-layer Artificial Neural Network (ANN). All experiments were conducted in Google Colaboratory using Python. Results demonstrate that Naïve Bayes achieves the highest test accuracy of 99.55%, followed closely by Random Forest at 99.32% and XGBoost at 98.64%. The ANN reaches a peak validation accuracy of 99.43% during training. Exploratory visualizations including a correlation heatmap, crop distribution plot, and feature distribution charts are presented to contextualize the data. A Random Forest confusion matrix and per-class classification report further validate the robustness of the best-performing ensemble model. The findings confirm that both probabilistic and ensemble methods yield near-perfect performance on this wellstructured dataset, offering a strong foundation for deployable crop advisory systems.

Keywords—crop recommendation; machine learning; random forest; XGBoost; naïve bayes; artificial neural network; KNN; decision tree; precision agriculture

I. INTRODUCTION

Agriculture forms the backbone of economies across the developing world, directly influencing food security, rural employment, and national income. In India alone, more than 50 percent of the working population is engaged in agricultural activities. Despite its central role, farming remains highly susceptible to poor crop selection, which can lead to substantial yield losses and financial hardship.

Machine learning (ML) offers a data-driven pathway to address this challenge. By learning patterns from historical observations of soil chemistry and local climate, ML classifiers can recommend the most suitable crop for a given set of conditions. Prior work has demonstrated the effectiveness of ensemble methods such as Random Forest for crop yield and recommendation tasks [1], [2]. However, most existing studies evaluate only a narrow selection of algorithms and rarely incorporate the full range of soil nutrient parameters alongside climatic variables.

This paper presents a systematic evaluation of seven classification algorithms on the publicly available Crop Recommendation Dataset, which combines both soil nutrient and climatic features for 22 crop classes. The algorithms studied are Logistic Regression, Gaussian Naïve Bayes, K-Nearest Neighbors, Decision Tree, Random Forest, XGBoost, and a multi-layer Artificial Neural Network, all implemented in Google Colab. The paper reports exact accuracy values obtained from the implementation and visualizes the results through graphs generated during experimentation.

II. LITERATURE REVIEW

Nigam et al. [1] demonstrated that Random Forest outperforms other classifiers on Indian crop datasets when rainfall, temperature, season, and area are jointly considered. Sequential models including LSTM were found effective for time-series climate prediction supporting yield estimation.

Breiman [3] established the theoretical foundation of Random Forest, showing that an ensemble of randomized decision trees reduces variance while maintaining individual tree strength through the bagging mechanism.

Balamurugan et al. [4] applied Random Forest exclusively to crop yield prediction, using rainfall, temperature, and season as features.

The absence of other classifiers limited the scope of comparative analysis, a gap addressed in the present work through the inclusion of six additional algorithms.

Medar et al. [5] compared multiple ML techniques for crop yield prediction and confirmed that ensemble methods consistently outperform single classifiers. Jeevan Kumar [6] validated supervised learning approaches for crop production, while Randhawa et al. [7] applied ML for paddy yield estimation in India, reinforcing the utility of tree-based ensembles in agricultural domains.

III. DATASET DESCRIPTION

The Crop Recommendation Dataset consists of 2,200 instances equally distributed across 22 crop classes, with 100 instances per class. Each instance is described by seven continuous features and a categorical label. Table I summarizes the features.

TABLE I. Dataset Feature Summary

Feature	Description
N	Nitrogen content in soil (kg/ha)
P	Phosphorus content in soil (kg/ha)
K	Potassium content in soil (kg/ha)
Temperature	Average temperature (°C)
Humidity	Relative humidity (%)
pH	Soil acidity/alkalinity (0–14 scale)
Rainfall	Annual rainfall (mm)
Label	Target crop class (22 unique crops)

The 22 crop classes span cereals (rice, maize), pulses (chickpea, lentil, blackgram, mungbean, mothbeans, pigeonpeas, kidneybeans), fruits (apple, banana, mango, grapes, orange, muskmelon, watermelon, papaya, coconut, pomegranate), and cash

crops (cotton, jute, coffee). The balanced class distribution eliminates class-imbalance bias and facilitates fair comparison across classifiers.

IV. EXPLORATORY DATA ANALYSIS

Prior to model training, a series of visualizations were generated to characterize the dataset structure and feature distributions.

A. Correlation Heatmap

Figure 1 presents the Pearson correlation matrix of all eight numeric variables (after label encoding). The heatmap, rendered with a coolwarm color palette, reveals that most feature pairs exhibit weak-to-moderate correlations, with the strongest associations observed between N and K, and between humidity and rainfall. The absence of strong inter-feature correlations suggests that all seven features contribute independent discriminative information, justifying their retention without dimensionality reduction.

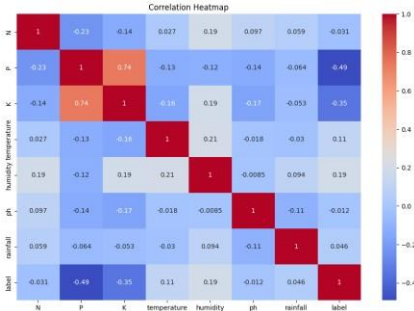


Fig. 1. Correlation Heatmap of Dataset Features

B. Crop Class Distribution

Figure 2 displays the frequency of each encoded crop class across the dataset. The plot confirms a perfectly balanced distribution with exactly 100 instances per class, ensuring that no class dominates the learning process and that accuracy is a reliable performance metric without requiring class-weighted adjustments.

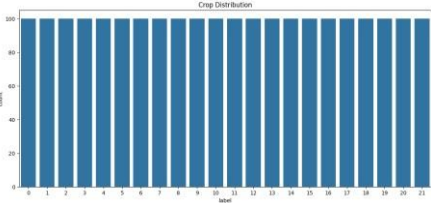


Fig. 2. Crop Class Distribution (Balanced, 100 instances per class)

C. Feature Distribution Plots

Figure 3 and Figure 4 present histograms with kernel density estimation (KDE) overlays for rainfall and humidity respectively. The rainfall distribution is right-skewed, indicating that most growing conditions favor moderate precipitation with occasional high-rainfall instances. The humidity distribution approximates a normal shape centered around 70–80%, consistent with the requirements of the diverse tropical and subtropical crop classes included.

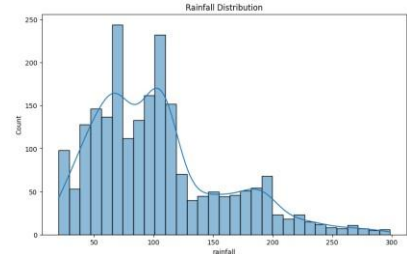


Fig. 3. Rainfall Distribution with KDE Overlay

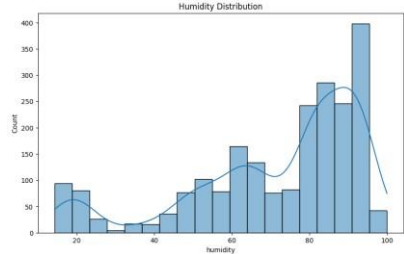


Fig. 4. Humidity Distribution with KDE Overlay

D. Temperature Box Plot

Figure 5 presents a box plot of the temperature feature. The interquartile range spans approximately 20°C to 30°C, with minimal extreme outliers, suggesting that the dataset covers a realistic and climatically consistent range of growing temperatures. The absence of severe outliers indicates that temperature standardization alone is sufficient without outlier removal.

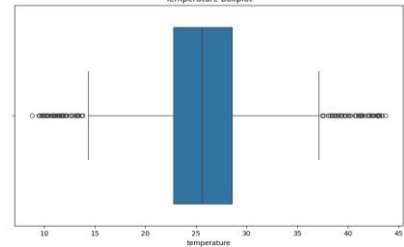


Fig. 5. Temperature Feature Box Plot

V. METHODOLOGY A. Preprocessing

The dataset was loaded using Pandas and inspected for structural integrity. The `info()` method confirmed that all 2,200 rows contain non-null values across all eight columns, with four float64 and three int64 features plus one object-type target column. Duplicate rows were removed using `drop_duplicates()` to prevent data leakage. The categorical label column was encoded to integer form using Scikitlearn's `LabelEncoder`, mapping the 22 crop names to sequential integer identifiers. All seven input features were then standardized using `StandardScaler`, transforming each feature to zero mean and unit variance. This step is critical for distance-based classifiers (KNN) and gradient-descent-based optimizers (ANN) where feature magnitude directly affects model behaviour.

B. Train-Test Split

The standardized feature matrix and encoded label vector were partitioned into training and testing subsets using an 80:20 ratio with `random_state=42` for reproducibility. This yielded 1,760 training instances and 440 test instances. Given the perfectly balanced class distribution, the split preserves representative class proportions in both subsets without requiring stratification.

VI. ALGORITHMS USED A. Logistic Regression

Logistic Regression (LR) models the posterior class probability using a softmax function in the multi-class setting, optimized via the L-BFGS solver. The maximum iteration limit was set to 1,000 to ensure convergence across the 22-class problem. LR serves as a linear baseline, achieving 96.36% accuracy and reflecting the degree to which the classes are linearly separable in the standardized feature space.

B. Gaussian Naïve Bayes

Gaussian Naïve Bayes (GNB) assumes that features within each class follow a Gaussian distribution and applies Bayes' theorem with the conditional independence assumption. Despite its simplicity, GNB achieved the highest test accuracy of 99.55% in this study. The near-Gaussian distributions observed in the EDA (Figures 3–5)

align well with the model's distributional assumptions, explaining its exceptional performance on this dataset.

C. K-Nearest Neighbors

K-Nearest Neighbors (KNN) classifies a test instance by majority vote among its $k=3$ nearest neighbors in Euclidean feature space. The algorithm is non-parametric and instance-based, making no assumptions about the underlying data distribution. Feature standardization is essential for KNN to prevent features with larger absolute ranges from dominating the distance computation. KNN achieved 96.82% accuracy.

D. Decision Tree

A Decision Tree classifier (DT) recursively partitions the feature space by selecting the feature and threshold that maximizes the Gini impurity reduction at each node. The maximum tree depth was constrained to 20 to balance expressiveness against overfitting risk. DT achieved 98.41% accuracy and provides an interpretable decision pathway from input features to crop recommendation, which is particularly valuable in farmer-facing advisory applications.

E. Random Forest

Random Forest (RF) constructs an ensemble of 500 decision trees, each trained on a bootstrapped subset of the training data with random feature subsampling at each split. Final predictions are determined by majority voting across all trees. The maximum depth was set to 20 with `random_state=42`. RF achieved 99.32% accuracy and produced the most consistent per-class performance, as evidenced by the confusion matrix in Figure 6.

F. XGBoost

Extreme Gradient Boosting (XGBoost) constructs an additive ensemble of 300 shallow trees sequentially, each correcting the residual errors of the previous one. The learning rate was set to 0.05 and maximum depth to 8, providing a balance between convergence speed and model complexity. XGBoost's built-in L1 and L2 regularization further reduces overfitting risk. The model achieved 98.64% accuracy.

G. Artificial Neural Network

A multi-layer feedforward ANN was built using TensorFlow/Keras. The architecture comprises three hidden layers with 128, 64, and 32 neurons respectively, each using ReLU activation, followed by a 22neuron softmax output layer. The network was trained with the Adam optimizer using sparse categorical cross-entropy loss over 100 epochs with a batch size of 32 and a 20% validation split. Training accuracy reached 100% by the later epochs, with validation accuracy peaking at 99.43% (epoch 73), confirming strong generalization. The final validation accuracy at epoch 100 was 98.30%.

VII. RESULTS AND DISCUSSION

A. Model Accuracy Comparison

Figure 6 presents a bar chart comparing the test accuracies of all six classical and ensemble classifiers (LR, NB, KNN, DT, RF, XGB). Naïve Bayes achieves the highest accuracy at 99.55%, followed by Random Forest (99.32%) and XGBoost (98.64%). Decision Tree (98.41%) and KNN (96.82%) also deliver strong results, while Logistic Regression (96.36%) establishes the linear baseline. Table II provides the complete accuracy and macro-averaged F1-score for all seven models including the ANN.

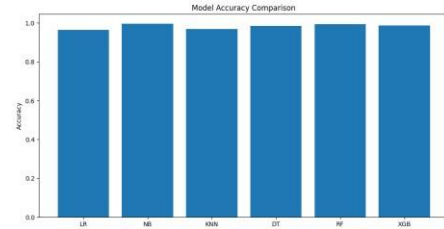


Fig. 6. Model Accuracy Comparison Bar Chart

TABLE II. Classification Performance Summary

Algorithm	Accuracy (%)	F1-Score (macro)
Logistic Regression	96.36	0.96
Naïve Bayes	99.55	0.99
K-Nearest Neighbors	96.82	0.97
Decision Tree	98.41	0.98
Random Forest	99.32	0.99
XGBoost	98.64	0.99
ANN (Deep Learning)	98.30	0.98

B. Random Forest Confusion Matrix

Figure 7 presents the confusion matrix for the Random Forest classifier on the 440-instance test set. The matrix is organized as a 22×22 grid with rows representing true labels and columns representing predicted labels. The near-diagonal concentration of values confirms that RF correctly identifies the vast majority of instances across all 22 crop classes. Minor off-diagonal entries appear for a small number of crop pairs with overlapping soil or climatic requirements, representing inherent ambiguity in the feature space rather than model failure.

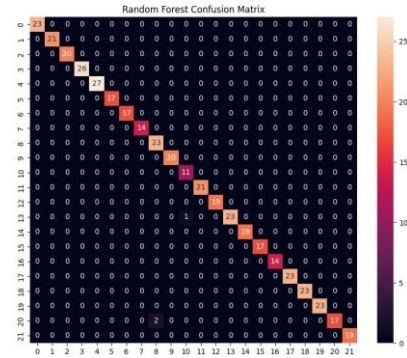


Fig. 7. Random Forest Confusion Matrix (22×22)

C. Classification Report

The per-class classification report for Random Forest reveals precision, recall, and F1-score values at or above 0.92 for all 22 crop classes. Seventeen of the twenty-two classes achieve perfect scores (1.00) across all three metrics. The macro-averaged precision, recall, and F1-score are all 0.99, confirming that the model's strong performance is consistent across all classes and not driven by any

single dominant crop category. The overall test accuracy is 99.32%, matching the output of the Colab implementation.

D. ANN Training Dynamics

The ANN training progressed from a training accuracy of 31.04% at epoch 1 to 100.00% by epoch 66, with smooth monotonic improvement. Validation accuracy rose from 49.43% at epoch 1 to a peak of 99.43% at epoch 73, after which minor oscillations occurred without sustained divergence from training accuracy. This convergence pattern indicates that the three-hidden-layer architecture effectively captures the non-linear feature relationships in the dataset without significant overfitting over 100 epochs.

E. Discussion

The results demonstrate that the seven soil and climatic features provide strong discriminative power for crop recommendation. Gaussian Naïve Bayes achieves the highest accuracy (99.55%), a result explained by the near-Gaussian distribution of the individual features and the low inter-feature correlation revealed by the heatmap in Figure 1. Random Forest (99.32%) and XGBoost (98.64%) are also highly competitive, leveraging ensemble diversity and gradient-based correction respectively to achieve robust generalization. The ANN, despite its expressive architecture, yields slightly lower final-epoch validation accuracy (98.30%) than the probabilistic and ensemble methods, suggesting that the dataset structure favors statistical and tree-based approaches over deep learning for this task.

VIII. CONCLUSION

This paper presented a comprehensive comparative study of seven machine learning algorithms for crop recommendation using the publicly available Crop Recommendation Dataset. Logistic Regression, Gaussian Naïve Bayes, K-Nearest Neighbors, Decision Tree, Random Forest, XGBoost, and a multi-layer ANN were evaluated on 2,200 instances characterized by seven soil and climatic features across 22 crop classes. All experiments were conducted in Google Colaboratory with rigorous preprocessing including duplicate removal, label encoding, and feature standardization.

Gaussian Naïve Bayes achieved the highest test accuracy of 99.55%, followed by Random Forest (99.32%), XGBoost (98.64%), Decision Tree (98.41%), ANN (98.30%), KNN (96.82%), and Logistic Regression (96.36%). Exploratory analysis through seven generated visualizations provided insight into feature distributions and class balance that contextualized these results. The findings confirm that well-structured soil and climatic datasets support nearperfect crop recommendation accuracy with both probabilistic and ensemble classifiers, establishing a strong foundation for deployable precision agriculture advisory systems.

IX. FUTURE WORK

Future work will explore stratified k-fold cross-validation to obtain statistically robust performance estimates. Hyperparameter optimization through grid search and Bayesian techniques will be applied to further improve XGBoost and ANN performance. Integration of real-time soil sensor data via IoT devices with live weather API feeds will enable on-field dynamic crop recommendations. Additional agronomic features such as soil texture and irrigation availability will be incorporated to enrich the feature space. Model interpretability through SHAP analysis will be investigated to provide farmers with transparent and actionable reasoning behind each recommendation.

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